EXECUTIVE PORTFOLIO BRIEFING · PROJECT 3

Customer Support Operations, NLP Triage & BI Command Center

Tech Stack: Python (Pandas, Regex, TextBlob, Scikit-Learn) · MySQL · Power BI (DAX, Custom SVG & JSON Theme)

7.79K	230 (2.95%)	3.00 / 5.0	5.65%	5.23K (67.2%)
USABLE TICKETS	URGENT RISKS (NLP)	AVG CSAT SCORE	AVG NLP SENTIMENT	ACTIVE BACKLOG

1. Executive Overview & Data Hygiene Funnel

SCOPE & PIPELINE

Customer support teams are routinely overwhelmed by unstructured text complaints, causing high-risk legal or hardware failures to sit buried in manual queues. This end-to-end analytics project ingests **8,469 raw historical support tickets** (17 attributes) in Python, enriches them with Natural Language Processing (NLP) sentiment and urgency flags, migrates 100% of raw records into MySQL, and applies strict Power Query data hygiene in Power BI to establish a validated operational baseline of **7,790 usable tickets (7.79K)**.

2. Phase 1 & 2: NLP Text Triage & Machine Learning Diagnostics

PYTHON & AI

- **Fixing the "Word Smashing" Regex Trap:** Initial punctuation stripping concatenated words inside dynamic template tags (e.g., `{product_purchased}` became `productpurchased`). Re-engineering the Regex cleaner to substitute non-letters with spaces and compress multi-space sequences preserved clean word boundaries for NLP tokenization.
- **Automated Sentiment & Urgency Triage:** Scored all descriptions via `TextBlob` (−1.0 to +1.0 polarity, averaging **+0.0565 or 5.65%** in the cleaned model) and deployed a rule-based scanner (`refund`, `cancel`, `lawsuit`, `lawyer`, `immediate`, `broken`, `unacceptable`) that isolated **247 raw urgent tickets** (**230 verified urgent tickets** post-cleaning).
- **Schema Audit & The 22% Random Forest Lesson:** Auditing the 2,769 closed tickets revealed that `Time to Resolution` was logged as a closing timestamp without a `Ticket Creation Date` column—making duration impossible to compute. Pivoting to a 100-tree **Random Forest Classifier** to predict 1-to-5 star CSAT from intake features yielded **22% accuracy** on 554 test tickets (vs. 20% random chance), proving mathematically that **intake sentiment does not dictate final satisfaction—agent resolution quality does**.

Pipeline Stage / Model Cohort	Record Volume	Key Metric / Output	Engineering & Business Finding
Stage 1: Raw Python & NLP Ingestion	**8,469 rows**	**247 Urgent (2.92%)**	Regex v2 word-boundary fix + TextBlob polarity scoring
Stage 2: Closed / Rated ML Subset	**2,769 rows**	**22.0% RF Accuracy**	Proved Intake ≠ Outcome; exposed missing creation timestamp
Stage 3: MySQL Database		**100% Import**	Defected Row 40 newline crash via strict

3. Phase 3 & 4: MySQL ETL & Power BI Command Center

BI & UI/UX

After resolving the Row-49 SQL newline crash in Python and bridging MySQL directly to Power BI (`'powerbi'@'%'`), a bespoke 16:9 dark-mode SVG wireframe (`cyber_slate_1280x720.svg`) and JSON theme (`Cyber_Slate_Theme.json`) were combined with DAX measures to surface live product, channel, and backlog bottlenecks.



Figure 1: Live Power BI Command Center displaying 7.79K usable tickets, 230 NLP urgent escalations, 3.00 Avg CSAT, 5.65% Avg NLP Sentiment, and 5.23K active backlog.

Command Center Visual	Primary Segment / Leader	Cleaned BI Metric	Operational Insight
1. The "Product Meltdown"	**LG Washing Machine**	**12 Urgent Tickets**	#1 hardware risk (vs. 8 each for Amazon, Canon, Samsung, Sony; 7 GoPro)
2. Channel Bottleneck	**Phone Support**	**5.07% Sentiment / 3.00 CSAT**	Lowest initial NLP sentiment (+0.0507) vs. Email (+0.0547) & Chat (+0.0542)
3. Ticket Status Share	Pending & Open Backlog	**67.15% (5.23K Tickets)**	33.97% Pending Customer Response + 33.18% Open vs. 32.84% Closed
4. Root Cause Treemap	Refund & Technical Issues	**1.61K & 1.60K Tickets**	Followed by Cancellation (1.55K), Billing (1.52K), & Product Inquiry (1.51K)
5. Live Escalation Queue	Ticket-Level Action Table	**230 Priority Rows**	Color-coded DAX Sentiment Category with sanitized complaint text

4. Strategic Recommendations for Leadership

ACTION PLAN

- **Immediate Appliance QA Audit (LG Washing Machine):** Investigate the **12 critical escalation tickets** on LG Washing Machines (50% above the runner-up hardware tier) to preempt flooding/damage claims.
- **Phone Queue De-Escalation & Backlog Triage:** Reallocate staffing to Phone Support (lowest intake sentiment at **+0.0507**) and automate self-service workflows for **Refund Requests (1.61K)** and